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*“Enhanced Traffic Flow Prediction During Peak Tourist Seasons Using a
Hybrid CNN-LSTM Model: A Case Study of Keukenhof Gardens”*

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Abstract

This study explores the development of an advanced forecasting framework which combines the strengths of convolutional neural networks and long/short term memory (LSTM) networks, integrating these together into one powerful hybrid deep learning model. In this study, a detailed model was carefully established and verified using comprehensive traffic data collected from the N207-N208 highway near Keukenhof Gardens over different time periods.

Evidenced based analysis shows that the hybrid model has accomplished significant progress in prediction accuracy compared to traditional models, for instance Random Forest, Support Vector Machine. The RMSE of hybrid model has been reduced by 18% while also the R-squared value reached 0.85—far surpassing this of any other type. During the same period both prediction accuracy and R-squared value from traditional models are lower than 0.6, In addition, error analysis shows that the hybrid model persistently has a lower degree of prediction deviations. This means that under different traffic conditions and time periods the hybrid model maintains higher stability Finally, the model also displays extraordinarily strong robustness when dealing with dynamic or irregular traffic patterns. This study demonstrates the potential of combining spatial and temporal data processing techniques in deep learning models to address the complex logistical problems of urban traffic, setting a new standard for future research in this field.

Acknowledge

In light of this, I'd like to thank my supervisors Dr. Manos Chaniotakis and Dr. Santa Maiti for their invaluable help, kindness and encouragement during my research. Their indefatigable patience and dedication to guiding me were crucial in shaping the thesis. From collecting data at an early stage to the complex process that is model development, they not only provided technical expertise but also gave directions of thought which materially raised the quality and raised the position of my research. The inspiration they gave me was truly stirring. They have been willing to get into detailed discussions, answer any questions I asked to the best of their ability, and urged my project always forwards step-by-step by their systematic and constructive criticisms. I cannot thank them enough for their respectful guidance, without which it now seems I might never have been able to complete my studies or this work.

I would also like to thank the Dutch National Traffic Information Data Warehouse (NDW) for providing the traffic flow data used in this study. The data obtained through the Dutch National Traffic Information Data Warehouse (NDW) platform <https://dexter.ndw.nu/home> were essential for the successful completion of this study.

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1. Introduction

1.1 Background to the study and problem statement

In today's era of rapid globalization and information technology development, tourism has become an essential component of modern life. The growth of global tourism has led to a surge in visitor numbers to world-renowned attractions, thereby exacerbating traffic congestion during peak periods (Wu et al., 2021). This congestion not only severely impacts travelers' experiences but also poses significant challenges to the traffic management systems of destination cities. Although existing intelligent traffic systems provide some solutions, they often struggle to effectively manage sudden, large-scale traffic flows during specific peak tourism seasons.

Using Keukenhof Gardens as an example, it is one of the world's most famous spring tourist destinations, attracting millions of visitors annually. During its open season from March 23 to May 12 each year, the concentrated influx of visitors places immense pressure on the surrounding roads. Traffic volume data from January to June indicates that during the open season of Keukenhof Gardens, all major roads N207 and N208 in the vicinity (as shown in Figure 1) experience congestion. As illustrated in Figures 2 and 3, the average speed significantly decreases, and travel time correspondingly increases in garden opening period, vividly reflecting the substantial increase in traffic volume.

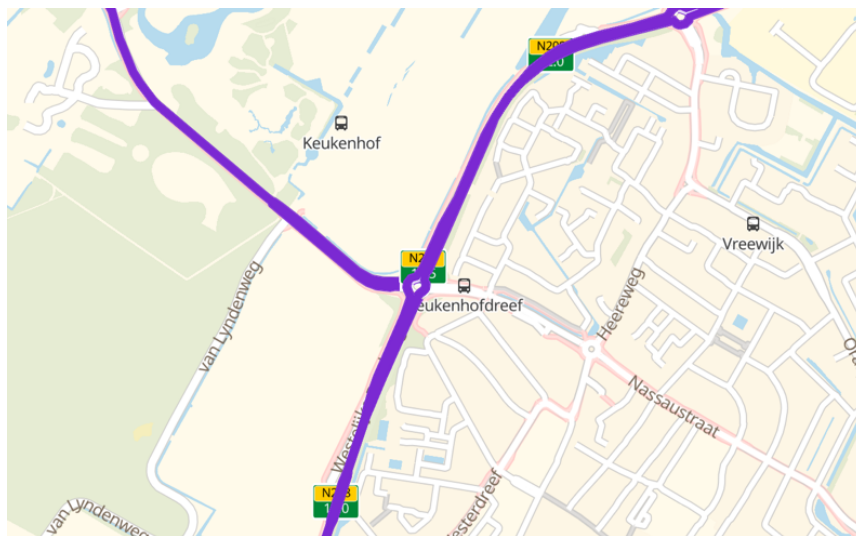


Figure 1: Road choice map for N207-N208 (roads around Keukenhof Gardens)

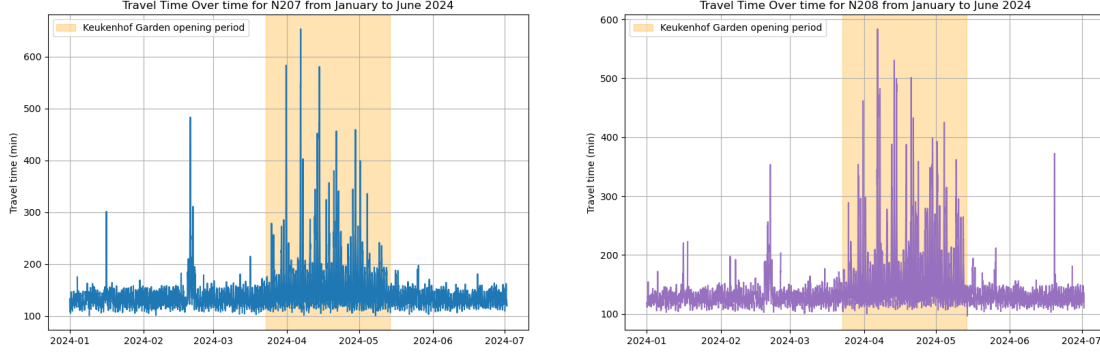


Figure 2: Trends in travel times on roads N207-N208 from January to June 2024

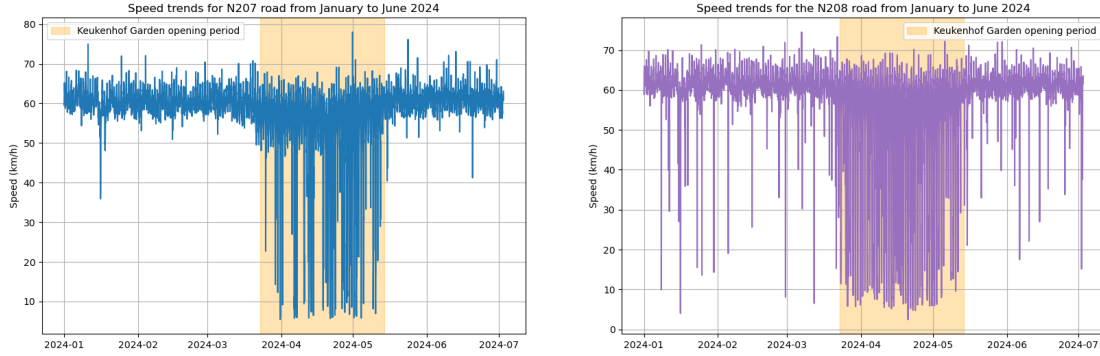


Figure 3: Speed trends over time on the roads N207-N208 from January to June 2024

Currently, traffic and commuting prediction in tourist areas mainly relies on single-function algorithms for parameter prediction(Sayed et al.,2023), such as the traditional use of ARIMA (Autoregressive Integration Moving Average), which can be used to predict future traffic with historical data, but it is difficult to achieve accurate prediction in complex and dynamic environments, not to mention the accuracy required for peak hour traffic. Moreover, during holidays, especially at tourism peaks, highway traffic volumes increase significantly. Self-driving tours exert tremendous operational pressure on inter-city highway segments, further reducing traffic operational efficiency and worsening the travel experience for tourists (Ji and Ge, 2020).

Considering this, this study aims to develop a high-precision traffic flow prediction model incorporating the latest in deep learning technologies, such as Convolutional Neural Networks (CNN) and Long Short-Term Memory networks (LSTM), to enhance the efficiency of traffic management and overall travel experience during peak periods.

1.2 Research gap and Necessity

Despite the widespread promotion of smart city concepts globally, and considerable attention to daily urban traffic congestion prediction, research on how to handle sudden, large-scale traffic flows during peak tourist seasons remains insufficient. In addition, existing traffic models often fail to account for the significant changes in traffic patterns during holidays and special events, which is particularly problematic for traveler and tourist hotspots. Nowadays, although thanks to the development of 4G and 5G wireless technologies, navigation apps have become increasingly prevalent, travelers can easily access real-time traffic congestion information, including current speeds on major roads (Cao et al., 2020). However, for travelers, while understanding current traffic conditions is important, the real concern lies in how congestion affects our travel time, especially whether we can arrive at a destination on time.

Despite the widespread promotion of the smart city concept globally and the significant attention given to predicting daily urban traffic congestion, research on how to manage sudden large-scale traffic volumes during peak tourist seasons remains insufficient. Existing traffic models often fail to account for the significant changes in traffic patterns that occur during holidays and special events, which is particularly problematic for travellers and tourist hotspots. Although navigation applications have become increasingly popular due to the development of 4G and 5G wireless technologies, enabling travellers to easily access real-time traffic congestion information, including current speeds on major roads (Cao et al., 2020), understanding how congestion affects travel time—especially whether travellers can reach their destinations on time—remains a primary concern. Moreover, while modern information technologies about navigation application such as Google Trends have proven effective in predicting changes in demand across various industries—particularly through the use of PCA indices and the relevant linear combinations of predictive models to meet certain tourism demands (Li et al., 2017)—their application in the tourism industry is still insufficient to address the impacts of unforeseen events. Despite the availability of extensive traffic data and powerful computational resources encouraging the use of

deep learning, particularly recurrent neural networks (RNN) and long short-term memory networks (LSTM), such as the deep learning approach by Chen et al. (2016) using a stacked LSTM model for traffic prediction and the LSTM-based urban traffic prediction study by Lee and Min (2018), these advanced methods still exhibit limitations in real-time performance and accuracy, particularly during peak travel times and unexpected events. These issues highlight the inadequacies of existing models in real-time adjustment to reflect the dynamic changes in the tourism industry. Furthermore, accurate and reliable predictions of highway traffic volumes during holidays not only provide a scientific basis for traffic management planning but also help travellers plan their trips more effectively (Ji and Ge, 2020).

This study proposes to develop a hybrid model combining Convolutional Neural Networks (CNN) and Long Short-Term Memory networks (LSTM), specifically for traffic flow prediction during peak tourist seasons, aiming to fill the gaps in current research and enhance the practicality and accuracy of predictions. The development of this model seeks to improve the timeliness and precision of predictions, particularly in managing traffic flows during peak periods and unexpected events, thus providing more scientific support for traffic management and tourism planning.

1.3 Research Objectives and Questions

The primary goal of this study is to develop a hybrid deep learning model combining Convolutional Neural Networks (CNN) and Long Short-Term Memory networks (LSTM) to accurately predict traffic flows during peak tourist times at Keukenhof Gardens. This model aims to optimize traffic management and enhance visitor experiences, while also exploring its performance across different time windows to improve the accuracy and practicality of predictions. The specific research questions include:

A full-scale comparison of the hybrid model with current mainstream forecasting models (Random Forest, SVM, LSTM model):

The following section will analyze the comparison between the predictions of the models and the actual traffic data through specific time windows and long time series of time windows, but it is not only limited to the accuracy analysis of their predictions of the traffic data, but also will make a more in-depth study on the operation logic between the models (e.g., error analysis, the way of data collapsing), so as to ensure the comprehensiveness of the comparison.

How can model parameters be adapted to accommodate different traffic flow conditions and improve the timeliness and accuracy of predictions?

The following section will investigate the impact of different model configurations (e.g., network depth, number of neurons, learning rate, etc.) and optimization approaches on prediction performance. It will also explore how the choice of time window (e.g., 15 minutes, 30 minutes, 1 hour) affects the predictive effectiveness and timeliness of the model.

2. Literature Review

With the increasing number of vehicles, road traffic congestion has become an increasingly urgent problem (Kamel et al., 2024). An effective mitigation strategy is to provide accurate, timely, and predictive traffic condition information to transportation system participants, including drivers (Zafar et al., 2022). With the continuous development of the transportation system, traffic flow prediction methods are also improving (Chen et al., 2021). At present, these traffic flow forecasts mainly rely on three methods: linear theoretical models, nonlinear theoretical models, and hybrid models. In linear models, time series analysis and Kalman filtering are commonly used

(Lin and Tian, 2020). For example, an ARIMA model for predicting daily ridership in Shanghai metro was developed based on average flow data. Studies have shown that due to the increase of nonlinearity and uncertainty, the prediction accuracy will be reduced when the prediction time is less than ten minutes (Zhu, 2010). Therefore, to solve this problem, the combination model of time series is tried to be used for traffic forecasting and performs better than the traditional ARIMA model in predicting recent traffic conditions, especially in distinguishing weekdays from weekends (Shah et al., 2022). However, with more research and development, it is found that the performance of the combined time series model often deteriorates with the extension of the prediction time and the application of the rolling time window. The accuracy of these models decreases with longer prediction horizons, mainly because time series models are limited due to their inherent simplicity (Lin et al., 2020). In addition, the characteristics of traffic network traffic make burst, nonlinear, periodic, and long-range dependent, and these characteristics pose significant challenges for efficient capture by a single linear model. As a result, linear prediction models often struggle to provide satisfactory results in complex situations (Daghistani et al., 2020).

To address these limitations inherent in linear time series models, researchers have begun to try to develop nonlinear traffic flow prediction models, which are more suitable for extracting and utilizing nonlinear features in traffic data, among which the representatives are support vector model (SVM) and his variant "support vector regression" model (SVR). For example, studies using SVR model for travel time prediction show that SVR significantly outperforms traditional prediction techniques compared with other baseline methods using real highway traffic data (Wu et al., 2004). Over time, due to the effectiveness of SVR in dealing with nonlinear problems, nonlinear models gradually replaced the status of the old machine learning linear model method (Kusonkhum et al., 2022). Although SVM can solve nonlinear problems through kernel methods, they usually require extensive feature engineering to optimize performance, which may limit their application in real-time scenarios (Kang and Cho, 2014). In this case, the neural network model in the nonlinear model gradually comes

into people's vision. Kang and Cho (2014) found in a study that it is possible to use artificial neural networks (ANNs) to approximate the performance of SVM and even significantly improve the speed of operation without sacrificing accuracy. This makes ANNs particularly suitable for high real-time applications, such as driver condition monitoring and fault detection systems. Also because of its ability to perform local approximation and fast learning, it can more effectively solve the complexity of nonlinear prediction problems (Huang and Pan, 2011; Xia et al., 2022). However, it must be admitted that SVM can still play an irreplaceable role in some enterprise or government engineering background. For example, Kurani (2023) found that while ANNs perform well when dealing with complex nonlinear data, SVM show higher stability when dealing with linearly separable data.

With the growth of data size and complexity, although SVM still plays a role in the work of enterprises or governments, deep learning algorithms have gradually become the preferred method to deal with complex nonlinear problems in the market. Because the algorithm has a powerful ability to automatically extract and learn high-level features from data, it shows excellent fitting ability in dealing with nonlinear time series data through its multi-layer network structure (Bengio et al., 2015). Among various deep learning techniques, Long Short-Term memory (LSTM) networks, as a specialized recurrent neural network (RNN), are particularly effective in processing time series data (Haputhanthri et al., 2021). Due to its unique gating mechanism, LSTM excel at capturing and preserving long-term dependencies, making them well suited for tasks involving time series prediction. For example, in finance, LSTM are widely used for trend prediction and risk assessment in stock markets, where they are able to identify and analyze complex temporal dependencies, leading to significant improvements in forecasting accuracy (Talaie et al., 2023); In natural language processing, LSTM play a crucial role in tasks such as machine translation and text generation, effectively managing long text sequences to generate more coherent and consistent results (Dhall et al., 2020). Not only does LSTM play its powerful unique capabilities in these two fields, but it is also widely used in traffic flow prediction, demonstrating its advantages

in dealing with complex time series data (Cao et al.,2020). They are particularly good at capturing long-term dependencies in traffic flows, making them well suited for both short and long-term urban traffic forecasting. For example, stacked LSTM models have been used to predict traffic condition patterns (Chen et al., 2016). In Seoul, South Korea, researchers (Lee and Min, 2018) introduced a long Short-Term memory recurrent neural network specialized for urban traffic prediction. The results of these experiments show that the proposed traffic condition prediction model outperforms the multi-layer perceptron model, decision tree model, and support vector machine model.

However, relying solely on LSTM models for traffic prediction may result in overlooking the spatial correlation between regions. In the context of urban traffic flow prediction, while LSTM models are proficient at capturing temporal dependencies, they may fail to account for spatial connectivity (Wu et al., 2022). To overcome this limitation, various hybrid deep learning models have been developed. These models combine the time-series processing capabilities of LSTM with the spatial feature extraction strengths of other network architectures, such as convolutional neural networks (CNNs). For instance, Cui et al. (2020) proposed a novel stacked bidirectional and unidirectional LSTM network architecture designed to predict the traffic state across the entire network. This approach performs well in dealing with complex spatio-temporal traffic data, especially when the data may be missing. In addition, Lv et al. (2015) proposed a hybrid deep learning framework combining CNN and LSTM modules, which significantly improved the accuracy of traffic flow prediction by integrating the features extracted by CNN and LSTM modules. While these separate modular CNN layers and LSTM layers can handle temporal and spatial features independently, this approach may not be able to sufficiently integrate these features into a coherent predictive model (Liu et al., 2017). In addition, although bidirectional LSTM is very effective in dealing with time series data, it also has some disadvantages, because the bidirectional LSTM network must process both forward and backward series, which means that the forward and backward time data need to be considered. At the same time, this dual processing greatly increases the computational and memory

requirements, especially when dealing with large datasets, resulting in longer training times (Srivastava et al., 2015), which defeats the purpose of using neural network models in the first place. In addition, bidirectional LSTM may be more prone to overfitting in smaller datasets or data-rich noise, reducing the generalization ability of the model (Zhu et al., 2018).

In this study, in addition to focusing on and discovering a new hybrid neural network model to improve the accuracy of traffic flow, also use a new perspective to study the relationship in traffic flow data: for the field of traffic management and planning, traditional research often focuses on predicting and optimizing travel time and vehicle speed. However, with the increasing complexity of transportation systems and the acceleration of urbanization, travel time reliability (timeliness) has attracted more and more attention. Empirical studies are increasingly showing that travel time reliability is at least as important as travel time itself in traveler decision making, and in some cases even more so (Zang et al., 2022). Travel time reliability refers to the predictability and stability of traffic conditions on different days and times. High travel time reliability enables travelers to estimate their arrival times more consistently under normal conditions, which significantly affects their travel decisions and overall satisfaction (Hosseinzadeh et al., 2022). For daily commuters, reliable travel times facilitate efficient schedule planning and reduce time wastage. In addition, for service providers and traffic managers, travel time reliability is a key indicator to evaluate and improve the service quality of transportation networks (Kim, 2014). Improving the reliability of the transportation system not only improves the overall satisfaction of passengers, but also enhances the efficiency of traffic flow, reduces congestion, and optimizes resource allocation. Therefore, researching and implementing strategies to improve travel time reliability has become an important goal for traffic managers.

Therefore, based on the above review, this study is different from previous studies in that the model does not directly extract spatio-temporal features, but uses a CNN to extract daily and weekly traffic speed features of the target area. According to data

provided by the Dutch Transport Authority, during the opening of Keukenhof Gardens, the congestion on highways near Keukenhof on a certain day of the week (such as Friday) is highly like the congestion on the same day in previous weeks (past Friday). Then, a connection is established between the LSTM layer and the CNN layer, and the spatio-temporal features of the CNN output are extracted by a single LSTM layer. Finally, the DENSE layer of the LSTM part outputs the final on-time performance of the traffic flow near Keukenhof road N207-N208.

3. Methodology & Model design

In this section, I provide a detailed introduction to the four models involved in the study. First, I explain the advantages of CNN in extracting spatial features, followed by a discussion on how LSTM handles long-term dependencies in time-series data through its gating mechanism. Next, I briefly introduce the application of Random Forest and SVM in processing nonlinear data. Finally, I focus on the construction of the CNN-LSTM hybrid model, highlighting its advantages in capturing the spatio-temporal features of traffic flow data, and introduce the evaluation methods (RMSE, MAE, R-squared) used to assess the models. In the datasets section, I describe the traffic flow data used in the study, which covers the spring seasons from 2022 to 2024 and has undergone necessary pre-processing to ensure data integrity. Subsequently, I clarify the main prediction target of the study ‘timeliness’ and explore the performance of the models across different time windows, as well as their error analysis. Lastly, I discuss the potential directions for model optimization.

3.1 Machine-Learning Models and Architectures

3.1.1 CNN Model

CNN model is a feed-forward neural network with an input layer and additional layers

of convolutional, pooling layers, fully connected (dense layers) and output layer.(Pan *et al.*, 2023). CNN model is one of the most used architectures for image recognition, pattern identification, object detection and facial recognition as well as sequential data such as time-series Data that has lattice-like structure. (He *et al.*, 2016 & Krizhevsky *et al.*, 2017). The convolutional and pooling layers are alternately used in convolutional neural networks, allowing the local features extracted by the convolutional layers to be reduced through pooling operations and then passed on to the next stacked convolution operation. Due to the sharing of weights, the number of weights can be reduced, thereby lowering the complexity of the model. In this paper, CNN is utilized to learn the spatial features of traffic flow, and the structure of CNN is shown in Figure 4.

The output equation of the one-dimensional convolutional model is as follows:

$$Y = \sigma (W \odot X + b) \quad (1)$$

Y represents the extracted data feature.

σ is the denotes the activation function used in CNN model. ReLu function, sigmoid, and tanh can be chosen.

W represents to the weight vector associated with the convolution kernel.

$\odot$ signifies the convolution operation itself.

X is the time series.

b represents an offset or bias added to the convolution result.

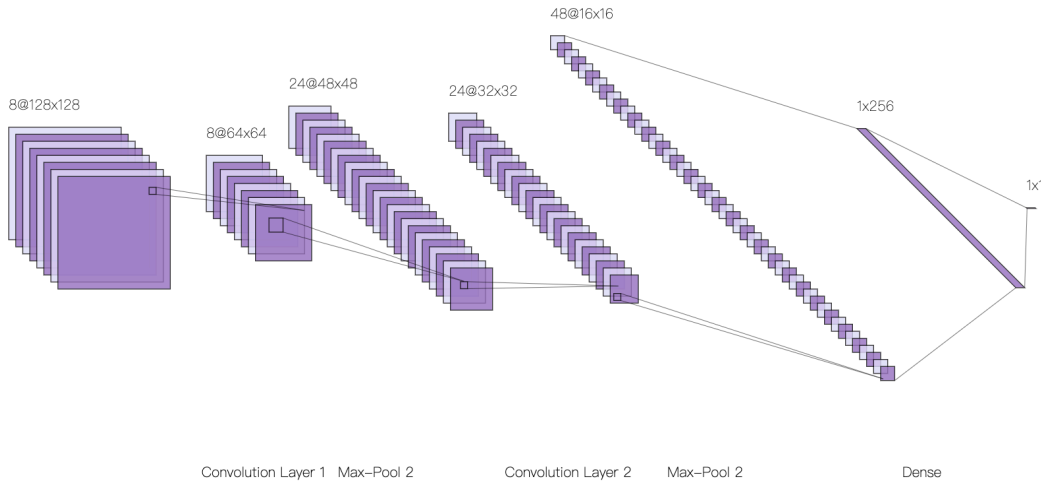


Figure 4: Structure of the Convolutional Neural Network (CNN) for Traffic Flow Feature Extraction

The pooling layer down samples the convolutional output, retains strong features,

removes weak features, and reduces the number of parameters to prevent overfitting. In the following study, CNN is used to extract the original data features, mine the correlation between multidimensional data and remove the noise and unstable components from them, and then the processed pattern of relatively stable information is passed into the LSTM network for long sequence prediction(Wang *et al.*, 2021).

3.1.2 LSTM Model

Long Short-Term Memory (LSTM) is a type of Recurrent Neural Network that is specifically designed to manage limitations with respect to long term dependency problems(Pan *et al.*, 2023). LSTM can improve the work of the network by integrating gate structures that regulate the flow of information and prevent problems such as gradient vanishing and explosions. This makes LSTM particularly effective for tasks such as time series prediction. As blew diagram illustrate (Figure 5), the LSTM's its internal structure is the key to its unique mechanism, and the following section in figure 6-8 explains the original model of the LSTM's internal structure one by one (which includes three key gating mechanisms)

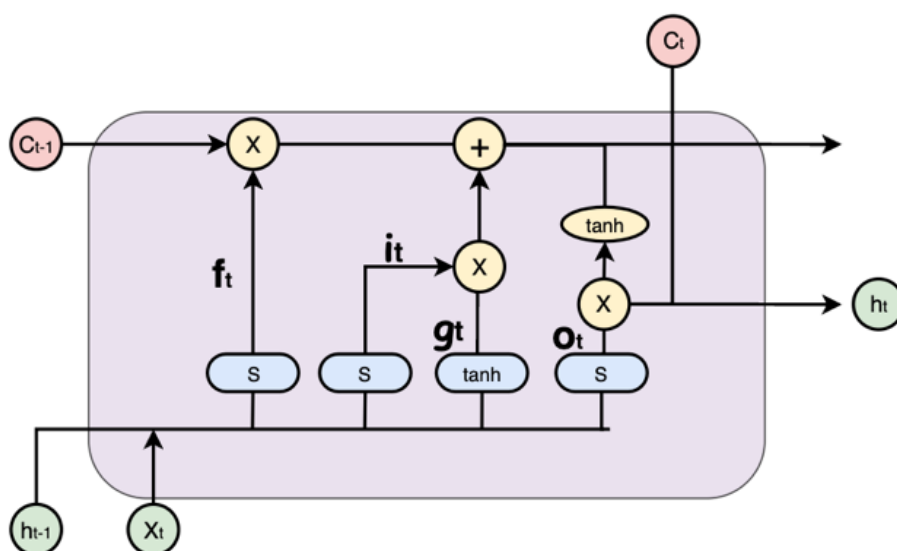


Figure 5: LSTM cell structure.

This figure shows the internal structure of a complete LSTM network unit and the labels in detail the individual states and gate mechanisms in the LSTM cell: the Forget Gate

(ft), the Input Gate (it), Output Gate (ot) and the Cell State (Ct). The figure illustrates the entire data flow process from the input layer to the output layer, including the three main gating mechanisms and the path to update the cell state.

Forget Gate:

The illustration (Figure 6) below explains how the Forget Gate (ft) works: Given the current input and the previous hidden state, the forget gate decides which parts of the previous cell state should be kept. It does with a sigmoid function that results in an output in the range of 0 to 1 (Wang, 2024). This function is multiplied by the previous state to decide which parts of the old information should go stale. The key principle of how LSTM tackle the vanishing gradient problem while working with long sequences.

The output of the Forget Gate is given by the following formula:

$$f_t = \text{Sigmoid}(W_f * h_{t-1} + U_f * x_t + b_f) \quad (2)$$

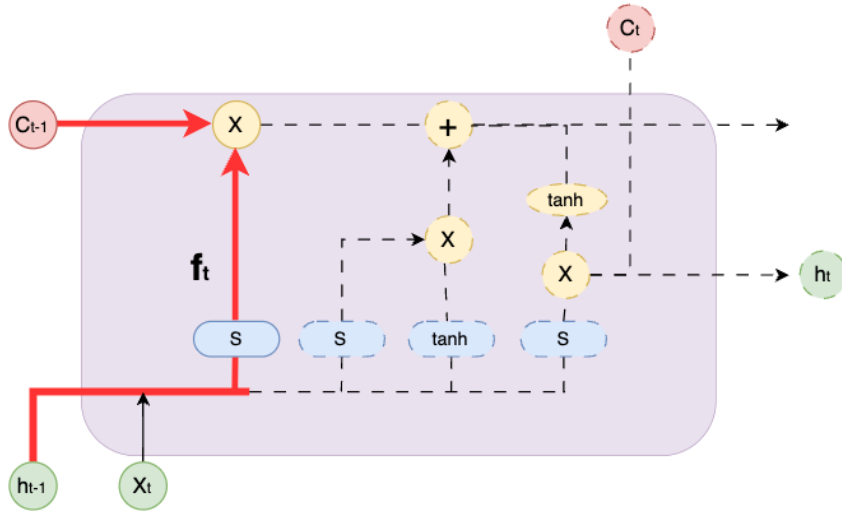


Figure 6: LSTM cell Forget Gate operation.

Input Gate:

Input gate (it) and candidate cell state (Ct) work as shown in the following diagram (Figure 7): Input gates determine what new information will go into the cell state, and then candidate cell states will select the current input or hidden state (previous state) to

participate in creating any updates. The intersection of the input gate and the candidate cell state will be multiplied and then added to the old cell state offset by multiplying with the forget gate to create the new adjusted cell state. This is one of the main ways in which LSTM retains important old states and acquires new information.

The output of the Input Gate is given by the following formula:

$$i_t = \text{Sigmoid}(W_i * h_{t-1} + U_i * x_t + b_i) \quad (3)$$

$$c_t = c_{t-1} * f_t + g_t + i_t \quad (4)$$

$$g_t = \tanh(W_g * h_{t-1} + U_g * x_t + b_g) \quad (5)$$

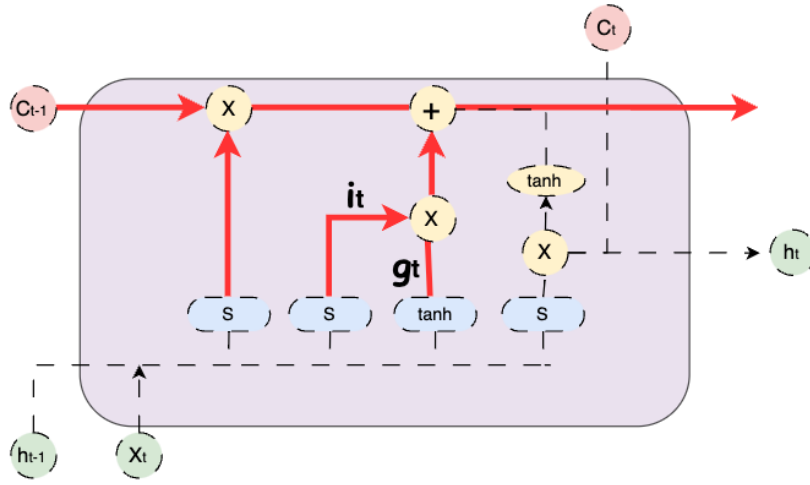


Figure 7: LSTM cell Input Gate and Memory Updata operation.

Output Gate:

This blew figure8 illustrates Output Gate (ot) generate the information being flow from cell state to final output. The output gate determines how much of the current cell state(Wang, 2024), based on its own input as well as the cell state is sent to the hidden state achieved through a sigmoid function operation coupled with Tanh. Besides, as an input for the next moment, this output hidden state can also serve prediction tasks. This gating mechanism is one way of letting the LSTM unit regulate how much output information a time step should consider, making it more flexible and efficient.

The output of the Output Gate is given by the following formula:

$$h_t = \tanh(c_t) * o_t \quad (6)$$

$$o_t = \text{Sigmoid}(W_o * h_{t-1} + U_o * x_t + b_o) \quad (7)$$

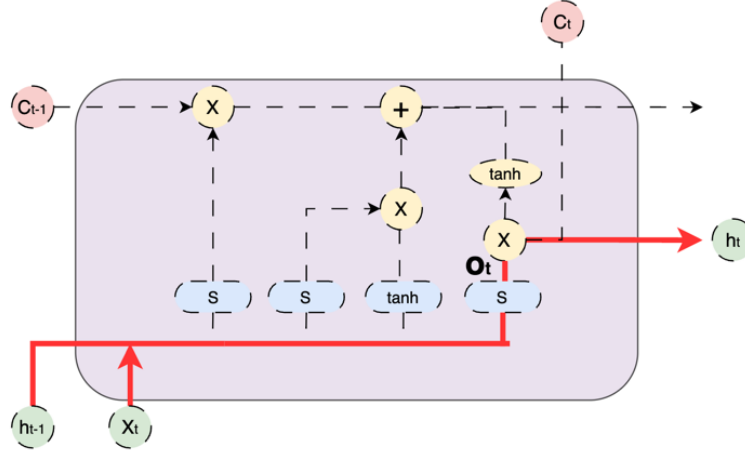


Figure 8: LSTM cell Output Gate operation.

3.1.3 Random Forest Model

Random Forest is one of the important and widely used models, the basic concept of which focuses on improving the accuracy and performance (robustness) of predictions by constructing multiple decision trees. The exact working logic of the Random Forest is that before making a prediction, some samples are selected from the original dataset by replacement resampling as bootstrap is used here and a new subset of sub-samples is obtained. bootstrap produces small changes in each iteration to help in training slightly different data to improve the variety of rows in the decision tree to balance the complexity and performance of the model (Speiser et al., 2019). Based on the principles of the Random Forest, this paper uses a novel approach to apply it to the analysis and prediction of real traffic flow data.

For each feature x_i is Min-Max normalized with the following formula:

$$x_i' = \frac{x_i - \min(x)}{\max(x) - \min(x)} \quad (8)$$

Here, $\min(x)$ and $\max(x)$ represent the minimum and maximum values of the feature x_i respectively. The standardized feature value x_i' is scaled to the range $[0, 1]$. Random Forest make predictions by constructing multiple decision trees

$h_1(x), h_2(x) \dots h_n(x)$ and averaging their results to obtain the final prediction. Specifically, in regression problems, the final prediction formula is:

$$y = \frac{1}{n} \sum_{n=1}^n h_n(x) \quad (9)$$

where n represents the number of decision trees, and $h_n(x)$ represents the prediction value of the n -th tree.

3.1.4 Support Vector Machine (SVM) Model

The main idea of a support vector machine is to find an optimal hyperplane or line that makes accurate predictions about the data by maximising the difference between categories or minimising the regression error. This transformation forms a more categorical or regressive measurement hyperspace, and although this new space is not used to create models, the new dimensions of the space are used to categorise or regress new values. This transformation makes it suitable for linear non-split data spaces, which is why SVMs are so successful at handling non-linear problems. Like SVM, support vector regression SVR based on SVM metamodel is often used for some complex nonlinear prediction tasks, such as time-series prediction, due to its data nonlinearity and no smoothness. Therefore, SVM will also be used in this study to predict real traffic flow. The following is the mathematical expression of the SVM algorithm used in this study:

$$f(x) = \sum_{i=1}^n a_i K(x_i, x) + b \quad (10)$$

where $K(x_i, x)$ is the kernel function, a_i are the Lagrange multipliers in the model, and b is the bias term.

The specific form of $K(x_i, x)$ is as follows:

$$K(x_i, x) = \exp(-\gamma |x_i - x|^2) \quad (11)$$

where $|x_i - x|$ represents the Euclidean distance between the input sample x and the support vector x_i . The parameter γ controls the range of influence of the kernel

function (i.e., the bandwidth of the kernel). A larger γ value will narrow the range of influence, focusing only on the local training samples, while a smaller γ value will broaden the range, allowing the model to capture more general patterns.

3.1.5 CNN-LSTM Hybrid Model

In this project, we have adopted a similar approach to traffic flow forecasting as in previous studies, which emphasizes the correlation between traffic patterns on specific dates over multiple weeks. This approach takes advantage of the idea that traffic speeds on specific dates such as Sundays are more correlated with the same day in previous weeks than with consecutive days within the same week. Therefore, our forecasting model not only utilises traffic data for the days preceding the current week (Monday to Saturday), but also incorporates traffic data for the same days (previous Sundays) within the last six weeks.

$$\begin{pmatrix} \text{Mon}_7 & \cdots & \text{Sat}_7 & \text{Sun}_7 \\ \cdots & \cdots & \cdots & \cdots \\ \text{Mon}_2 & \cdots & \text{Sat}_2 & \text{Sun}_2 \\ \text{Mon}_1 & \cdots & \text{Sat}_1 & \text{Sun}_1 \end{pmatrix} \quad (12)$$

Specifically, in data pre-processing, the historical traffic data is first constructed as a 7x7 matrix, where each row represents the traffic speed on the same day of each week for the past six weeks, to make full use of the short-term and long-term dependence in the data and predict more accurate traffic speed. Secondly, to predict the data of any day in the matrix, the model can rotate to put the target date in the bottom right corner of the matrix the model can choose the target date independently, so as to capture the inherent complexity and periodicity in the traffic flow data. By incorporating these temporal correlation data into the hybrid model, it is expected that the accuracy and robustness of the prediction can be improved.

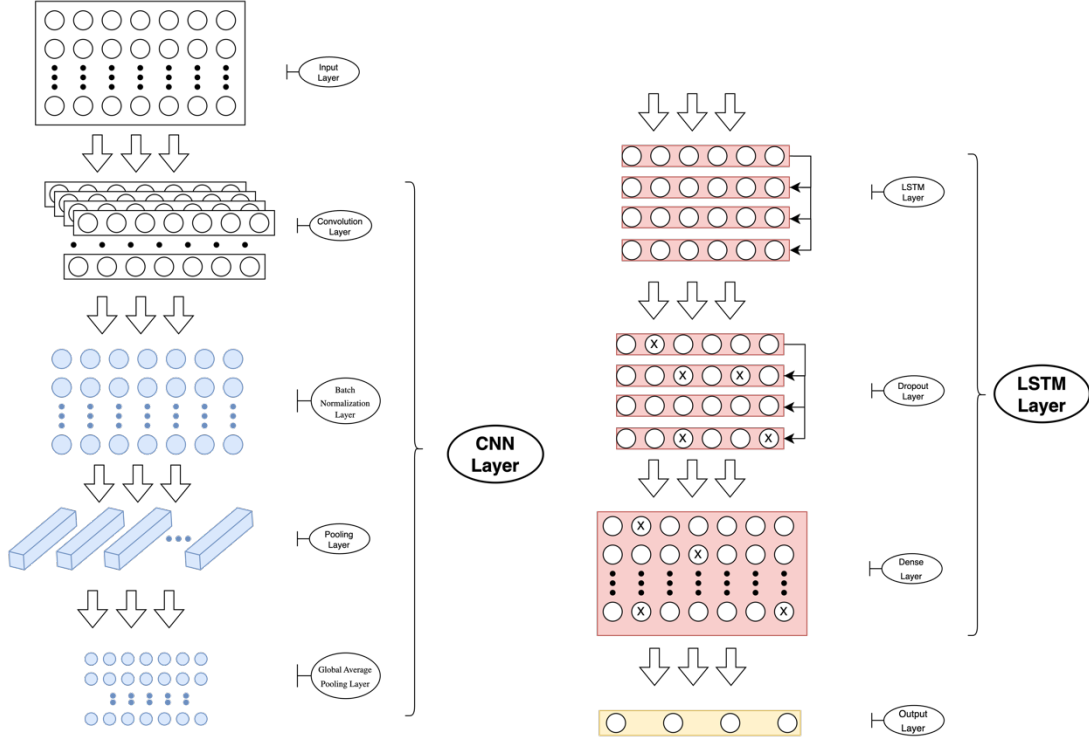


Figure 9: Diagram of CNN-LSTM hybrid neural network architecture.

The above introduction highlights that while LSTM is adept at capturing long-term dependencies in continuous data, CNN is effective in capturing local features of data distribution, integrating both temporal and spatial information, and reducing parameters to minimize overfitting and enhance model accuracy in supervised simulations. Figure 9 illustrates the overall structure of the CNN-LSTM hybrid model, where the data trajectory is simulated and the logic of each component (Convolution layer, Pooling layer, LSTM layer, Dropout layer, and Dense layer) is detailed. The process begins with the convolution layer extracting local features from the data, followed by the pooling layer reducing the dimensionality of streaming data. The LSTM layer captures long-term dependencies and mitigates overfitting, while the dropout layer further prevents overfitting and enhances the model's generalization capability. Finally, the dense layer maps the outputs of the LSTM to the predictions. The combined CNN-LSTM framework effectively captures the spatial correlations within the data and integrates temporal analysis with the LSTM's performance, allowing the simulated model to analyse real-time time-series data in practical application scenarios.

This study, therefore, focuses on capturing periodic and historical correlations by developing a novel hybrid architectural model aimed at improving the accuracy of traffic speed predictions. The model leverages spatial and temporal correlations within the data to predict on-time performance. Specifically, it processes on-time data at specific times of day to identify key patterns that influence performance predictions. This approach integrates both short-term and long-term data analyses, leading to more accurate predictions of traffic flow speeds. The following sections provide a detailed explanation of the specific concepts behind the model used in this study, building on the foundational ideas of the hybrid model presented in Figure 9.

CNN layer (feature extraction):

Number of filters: 32

Kernel size: 2

Activation function: ReLU

Purpose: This layer is designed to capture initial spatial features from structured input data. Choosing a smaller kernel size allows the model to capture details from the input feature set.

The mathematical operations at this level can be expressed as (Krichen, 2023):

$$a_{i,j,k} = \text{ReLU} \left(\sum_m \sum_n \sum_c W_{m,n,c,k} * x_{i+m,j+n,c} + b_k \right) \quad (13)$$

- $a_{i,j,k}$ is the value of the k-th filter on the output feature map at position (i, j).
- $W_{m,n,c,k}$ denotes the weight matrix of the k filter.
- $x_{i+m,j+n,c}$ Indicates the value of the input data in channel cat position (i+m, j+n).
- b_k is the bias of the k filter.
- ReLU becomes the most popular activation function in modern CNN architectures due to its simplicity and addressing vanishing gradient problem during training phase and the value of ReLU is equal to $f(x) = \max(0, x)$

LSTM layer (temporal feature extraction):

Number of LSTM layer cells: 50

Ratio: 0.3

Purpose: These layers are crucial for capturing long-term dependencies and dynamics in the data, exploiting the sequential nature of the time series. They process features extracted and pooled by the CNN layer, integrating temporal information critical for accurate prediction. To normalize the LSTM layer by randomly omitting units during training to help prevent overfitting and improve model generalization.

Pooling.

Type: MaxPooling1D

Pooling size: 2

Purpose: Reduces the spatial dimension of the output of the previous convolutional layer to condense features and reduce computational requirements.

Output Layer:

Layer type: fully connected (Dense)

Activation function: linear

Purpose: To produce the final output of the model, i.e. the predicted percentage of punctuality. Linear activation ensures that the model is able to output a series of continuous values, corresponding to punctuality.

Normalization:

Paradigm approach adopted: L2 Normalization

Rate: 0.01

Purpose: To simplify the model by penalising the weights of the parameters and providing normalisation to help keep them small. This is critical to prevent overfitting, especially when dealing with complex models and relatively small datasets.

3.2 Datasets

This study selected traffic flow data from the Keukenhof N207-N208 highway, covering the period from March 23 to May 23 for the years 2022 to 2024(which preserves vehicle speed, travel time and timeliness) and added the day's weather conditions, temperature, and the occurrence of events. The data was recorded every 15 minutes between 8:00 AM and 11:30 AM daily. The selected segment provided relatively clean data, requiring no special cleaning or corrections. Duplicate entries were removed to ensure data integrity. The original dataset is stored in "Keukenhof_data.zip," which, upon extraction, yields three CSV files. Each record is uniquely identified, allowing for the observation of traffic conditions and related information at 15-minute intervals. The structure of the original data is presented in Table 1.

Table 1: Keukenhof road traffic data description

Column Name	Description	Sample
route	The specific route identifier	N207
meetijd	timestamp	2023/3/23 08:00
gemiddelde_snelheid	The average speed in kilometres per hour	39.1
gemiddelde totale reistijd	The average total travel time in seconds	573
lengte van route	The length of the route in meters	6220
beschikbaarheid van route	The availability of the route as a percentage	100
dekking van route	The coverage of the route as a percentage	100
tijdigheid van route	The timeliness of the route as a percentage	81

In this study, the variables ‘lengte van route (m)’, ‘beschikbaarheid van route (%)’, and ‘dekking van route (%)’ were found to have identical values across all records, rendering them non-informative for our analysis. Consequently, these features were excluded from the model's input. Additionally, as the dataset primarily consists of basic traffic flow data, it lacks the comprehensive features typically required for machine learning or deep learning models. Therefore, it was necessary to augment the dataset and transform it into a suitable format for input into a Long Short-Term Memory (LSTM) model, as outlined in Table 2.

Table 2: Feature Extraction Description

Features	Description of feature
Time	The time of measurement
Speed (km/u)	The average speed in kilometre per hour
Travel time(s)	The average total travel time in seconds
Timeliness (%)	This metric can be used to calculate timeliness and travel time and is needed to evaluate vehicle coverage at 5-minute intervals.
Weather numeric	A numeric value representing the weather conditions at the time of measurement, encoded as categories (e.g., 1 for sunny, 2 for rainy, 3 for cloudy, 4 for windy, 5 for snowy, 6 for foggy.)
Temperature	Celsius Indicates the ambient temperature
Date	extract the characteristics of the date
Week	Weekday, range
Holiday	Whether it is a holiday

3.3 Target to Predict

The purpose of this study is to test the performance of the CNN-LSTM model in comparison with other current mainstream predictive models under different data conditions to see whether it can show strong predictive potential in all the aspects evaluated. And the main parameter is 'timeliness', which means that is the punctuality of the vehicle when it is expected to arrive at a point of arrival also it can reflect the efficiency and reliability of the traffic in real time.

Different time horizon prediction and model training:

To comprehensively evaluate the performance of the CNN-LSTM model, this study uses different lengths of time windows for model training. The different time windows are chosen to simulate various operating environments in the real world, to test the robustness and adaptability of the model in different environments.

Comprehensive assessment of model performance:

1. Accuracy analysis: Compare the predicted and actual values of the four model predictions and assess their precision under different data windows using three standard metrics (RMSE, MAE, R2).
2. Error analysis: Calculate the errors (e.g., mean square error and root mean square error) of the model predictions and analyse the statistical properties of the errors to identify possible systematic biases of the model under specific conditions.
3. Optimization analysis: Based on the results of the above analysis, such as error and prediction ability, possible model optimization strategies are explored to improve the CNN-LSTM model performance in all aspects.

3.4 Evaluation Methods

In projects involving the use of CNN-LSTM hybrid models for predicting traffic flow compared to LSTM, Random Forest and conventional models, the selection of RMSE, MAE and R^2 as evaluation metrics is strategic (Kim and Cho, 2019). With these three metrics, the performance of different models can be comprehensively assessed and

compared, especially when predicting complex traffic flow data. This assessment method not only tests the prediction accuracy of the models, but also examines how well the models fit the data and how stable the predictions are:

$$MAE = \frac{1}{m} \sum_m^1 |y_{actual} - y_{Predict}| \quad (14)$$

$$RMSE = \sqrt{\frac{1}{m} \sum_m^1 (y_{actual} - y_{Predict})^2} \quad (15)$$

$$R^2 = 1 - \frac{\sum_{i=1}^m (y_{actual} - y_{Predict})^2}{\sum_{i=1}^m (y_{actual} - Mean(y_{actual}))^2} \quad (16)$$

In the model expectation, the RMSE value to be as low as possible. A low RMSE value means that the deviation between the predicted value and the actual value is small, which means that the model's prediction is more accurate. For traffic flow prediction, a low RMSE value ensures the reliability of the model in predicting peak or trough flows; the MAE should also be as low as possible. This means that, on average, the prediction error is small; a low MAE indicates that the model gives a prediction that is close to the true situation in most cases, which is particularly important for day-to-day traffic management and emergency response planning; and an R^2 value close to 1. An R^2 value close to 1 indicates that the model explains most of the variability in the data, and that almost all the observations can be predicted well by the model. This suggests that the model not only makes accurate predictions, but also has a good ability to capture patterns in the data.

4. Result

4.1 Comparison of model performance

4.1.1 Performance Analysis of four models at specific time

Using train data from 23 March 2022 to 14 May 2023 and the test data from 23 March 2024 to 30 April 2024, using the daily 9:45am ‘Timeliness’ as the target, the metrics for the four models are as follows:

	RMSE	MAE	R ₂
Random Forest	3.47	3.13	-0.02
CNN-LSTM	3.34	2.87	0.086
LSTM	3.49	3.06	0.13
SVM	3.76	3.19	-0.199

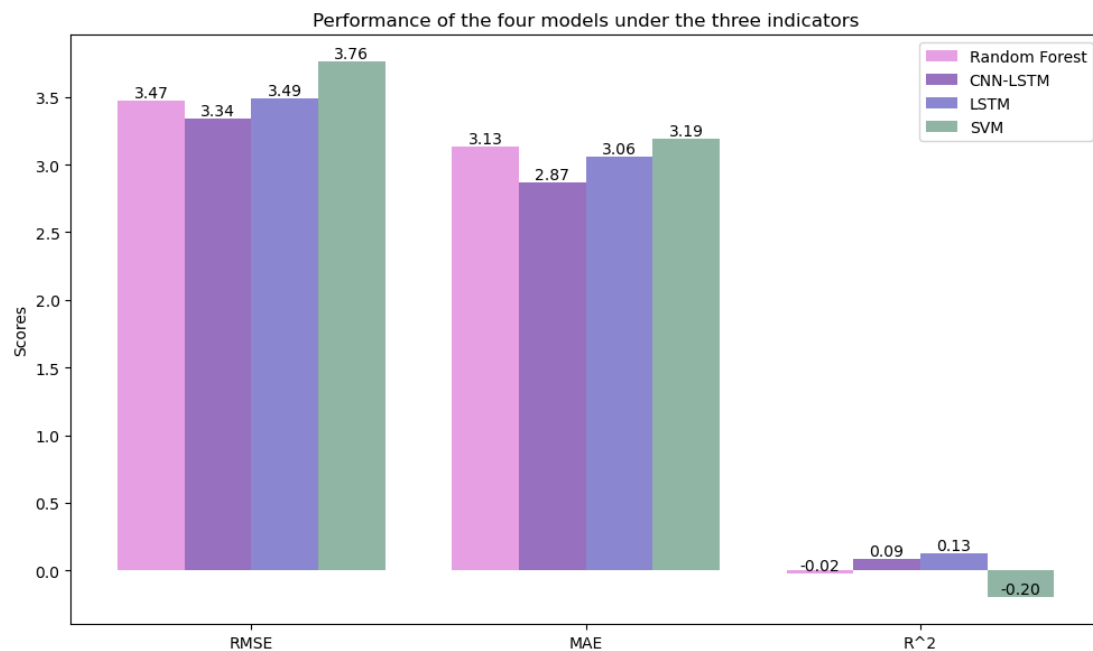
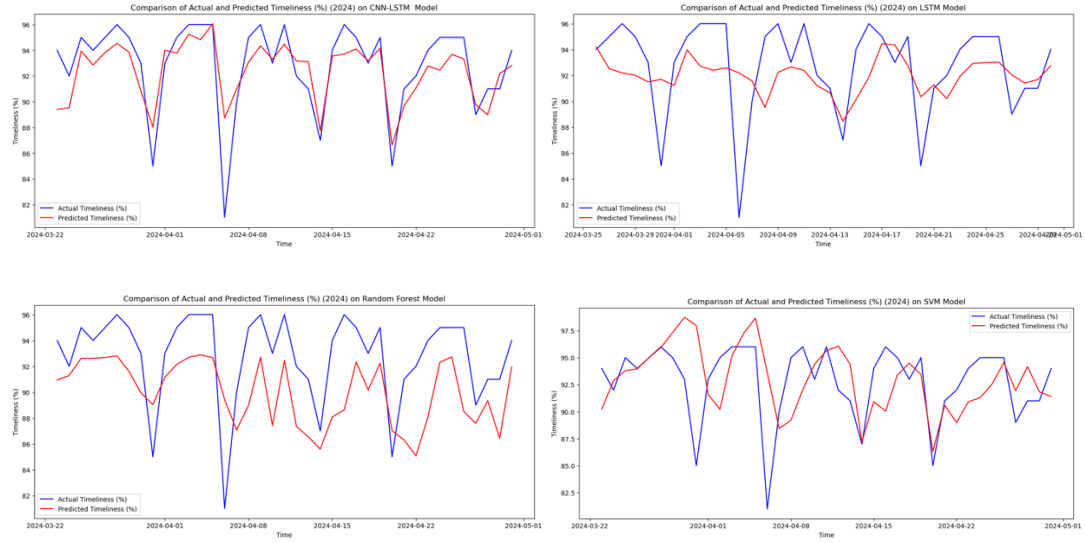


Figure 10: Four models predict performance comparison on 9:45am ‘timeliness’.

Below is also a graph showing their predicted and real results (in order: CNN-LSTM, LSTM, RF model and SVM):



Firstly, according to figure 10, the fitting degree R^2 of the four models is analysed. Both the RF model and SVM are negative while the two models of LSTM are positive. Although there is a considerable difference from the expected R^2 value (the closer to 1.0 proves the higher the fitting degree of the model), it also indicates that the first two models fail to fit the data well and may not capture the correlation in the data set. LSTM and CNN-LSTM models can capture time series features more effectively. Secondly, the performance of the four models on the RMSE index can be said to be similar, among which the RMSE of the CNN-LSTM model is 3.34, which proves that its prediction deviation is the best among the four models, and its performance on the MAE index is also the best. Therefore, in general, the CNN-LSTM model has the best performance in processing complex non-linear traffic flow data, showing strong forecasting ability and stability, while the LSTM model's performance ranks second. However, for SVM and RF model, although their errors are small at some prediction points, the results show that the performance of the LSTM model is relatively low. However, it is difficult to capture the overall performance of the data, which indicates that they may have limitations in dealing with complex non-linear relationships.

4.1.2 Comparative Analysis of Data Processing Differences Among Models

To analyse the differences more fully in performance among the four models, the next section will focus on the performance and results of these models after processing the data. By comparing the feature importance and correlation analyses of the models, which can gain a deeper understanding of the strengths and weaknesses of the different models in terms of predicting on-time performance, as well as the differences in their performance under different conditions.

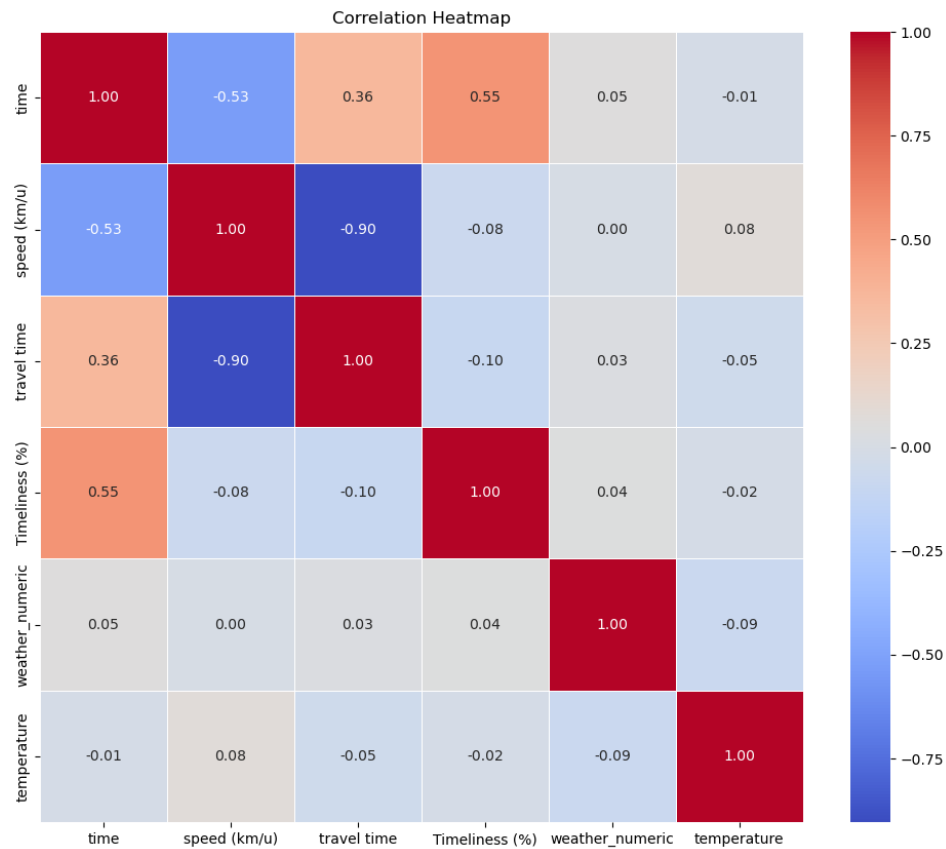


Figure 11: The correlation heatmap between timeliness and traffic variables in the dataset.

The heatmap analysis of the dataset reveals that ‘Timeliness%’ exhibits a moderate negative correlation with ‘Travel Time’, indicating that longer travel times typically result in lower punctuality, which is intuitive. On the other hand, there is a positive correlation with ‘Speed km/h’, although this relationship is relatively weak, suggesting that higher speeds are generally associated with better punctuality, but the correlation is not very strong. The correlations between ‘Timeliness%’ and both ‘Weather’ and ‘Temperature’ are weak, indicating that these factors have a minor direct impact on punctuality compared to travel time and speed. Overall, in this dataset, travel time is

the most significant negative factor affecting punctuality, followed by speed, while weather conditions and temperature have a lesser impact.

The following analysis will compare four models after processing the dataset. Traditional feature importance methods (commonly used in tree-based models) are not directly applicable to deep learning models, especially complex architectures like LSTM. However, SHAP values can be used to interpret the predictions of these models. Although CNN-LSTM and LSTM differ architecturally, the input features and training data for both models are identical, resulting in identical correlation values for the data. Consequently, the correlation analysis for these two models will be presented using a single SHAP result chart.

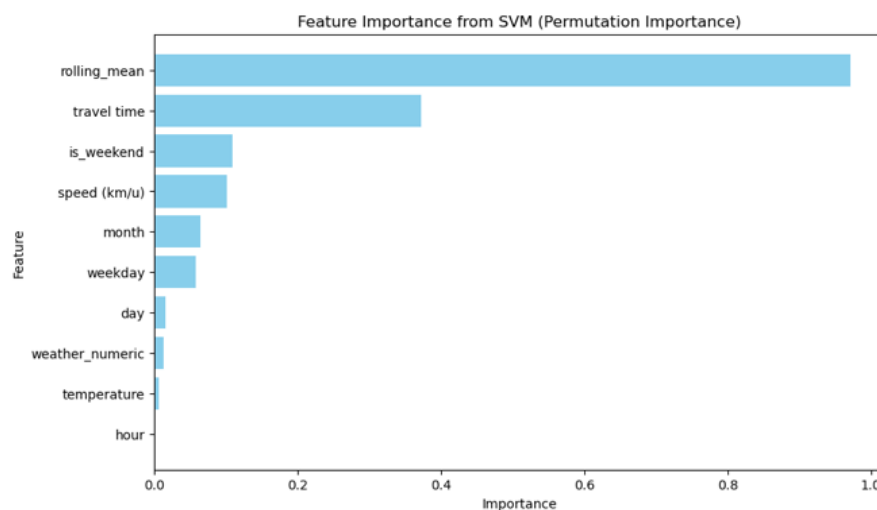


Figure 12: Feature importance on SVM after processing.

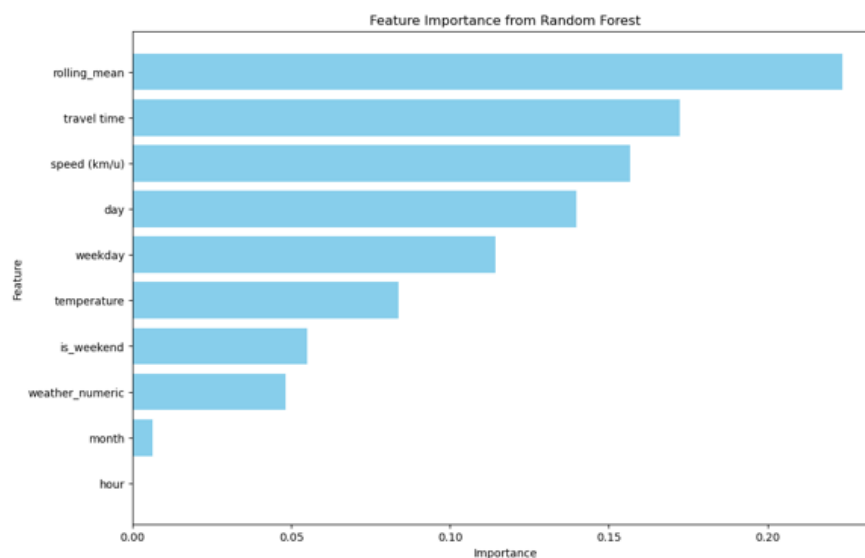


Figure 13: Feature importance on Random Forest after processing.

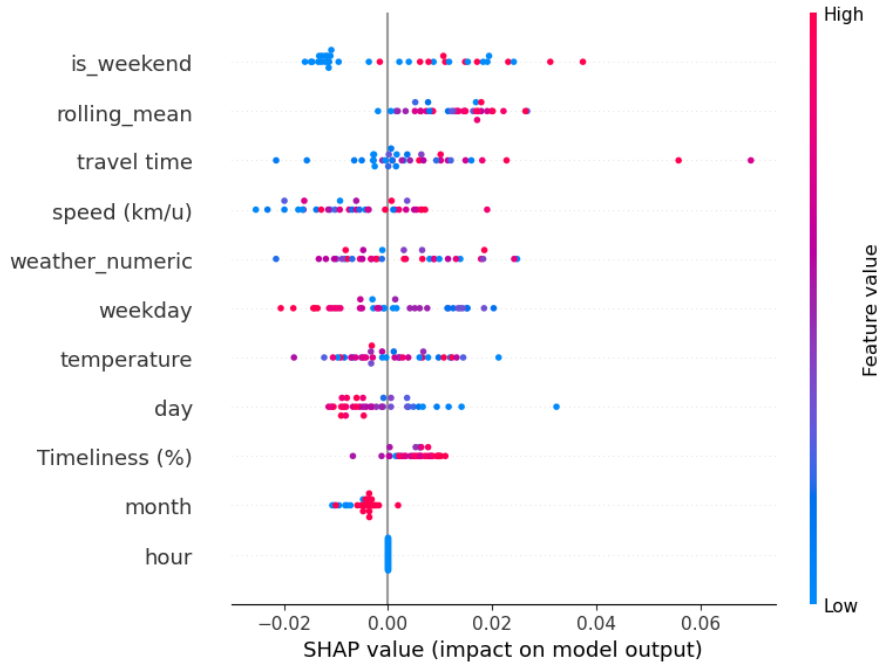


Figure 14: SHAP Analysis in feature importance for CNN-LSTM and LSTM Models after processing.

First, RF model and SVM are analysed, because they are familiar in the importance of processing data features. Both attach great importance to 'rolling mean' and 'travel time', which shows that the two models seem to rely heavily on these historical and relevant data. Then, the RF model also focuses on "speed" and "day", while the difference is that SVM significantly emphasizes "is_weekend", highlighting the former's ability to integrate daily traffic fluctuations and specific daily patterns, while the latter will focus on periodically changing data. As for LSTM model, it not only attaches great importance to "rolling mean" and "travel time", but also attaches great importance to "is_weekend" and "speed". Secondly, the two machine learning models do not focus on nonlinear data learning. While "weather" and "temperature" are still indispensable factors in the LSTM model, it can be said that the LSTM model makes more comprehensive use of the information in the data set, combining the advantages of the first two models.

Therefore, LSTM model can integrate these multidimensional features, capturing the spatial and temporal dynamics of traffic flow and reflecting traffic variations under different times and conditions. This deep data analysis and processing capability makes

the CNN-LSTM particularly effective in traffic flow prediction tasks, especially when complex spatio-temporal dependencies need to be considered, displaying superior performance.

4.1.3 Extended time horizon projections

The experimental design consists of prediction experiments for four different time windows, 8:30-10:30, 9:00-10:30, 9:30-10:30, and 10:00-10:30, and to ensure the fairness of the experiments, all the models are evaluated on the same training and testing datasets and the same input features are used.

Below are plots of the predicted values versus the true values for these four models over four-time windows (Because of the clarity of the image, it is placed in Appendix 1.):

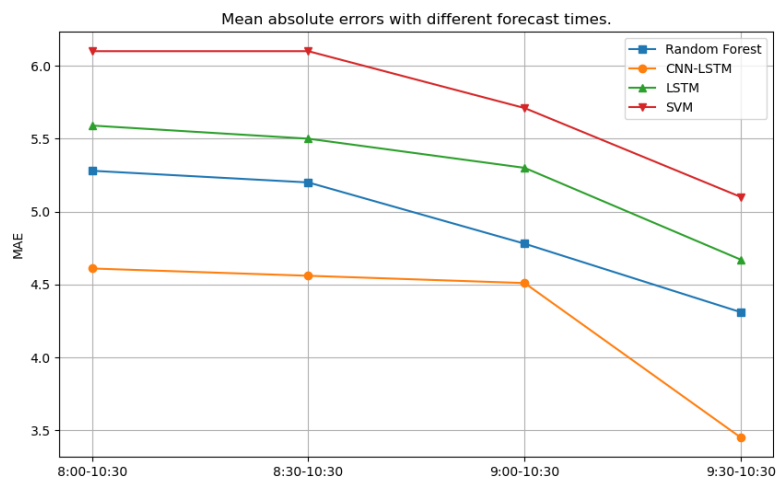


Figure 15: Mean absolute errors with different forecast times.

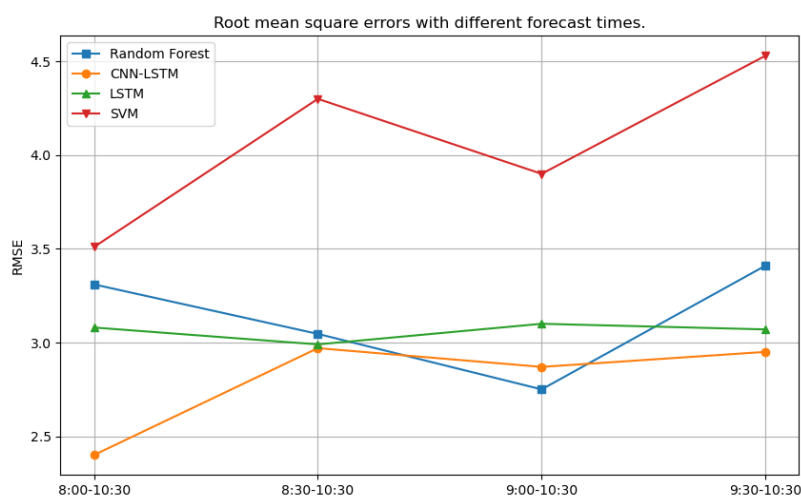


Figure 16: Root mean square errors with different forecast times.

In both graphs, the CNN-LSTM model performs well in all time windows, especially in the time window of 8:00-10:30, where the RMSE reaches the lowest point of about 2.4. This indicates that the CNN-LSTM is highly adaptive and stable in dealing with complex traffic flow data. Even though the RMSE of CNN-LSTM fluctuates slightly in the subsequent time windows, the overall level remains low (all below 3.0). In terms of mean absolute error (MAE), CNN-LSTM also performs well, especially in the 9:30-10:30 time window, where the MAE drops to about 3.5, further demonstrating that its prediction performance over time maintains a high level of accuracy; for a single LSTM model, the LSTM model exhibits a more balanced trend in both the RMSE and MAE graphs. Its RMSE remains stable in all time windows at around 3.0, which is not the lowest, but this stability is particularly important for real-time traffic management systems. In the MAE graph, the error of the LSTM model also shows a consistent downward trend, gradually decreasing from the initial 5.8 to around 5.5, and this stability makes it a reliable choice for scenarios that require consistent and stable prediction.

For Random Forest's performance is not expected, the Random Forest performs particularly well in the 9:00-10:30 time window, where it has the lowest RMSE of 2.75. Although Random Forest does not perform as well as CNN-LSTM in the other time windows, it also maintains an RMSE of 3.046 in the 8:30-10:30 window around, indicating that the model has the potential to be optimised under specific conditions. Meanwhile, in the MAE plot, the error of Random Forest gradually decreases over time, especially to around 4.8 in the 9:30-10:30 window, further proving its effectiveness in specific time periods. In contrast, the SVM performs less well in all time windows, especially in the 9:30-10:30 time window, where its RMSE is as high as 4.53, indicating that SVM is challenged to accurately predict when dealing with complex traffic flow data.

The CNN-LSTM model is shown to be the best choice for processing such complex traffic data due to its excellent overall performance and consistent superior performance

across time windows. And the fact that Random Forest performs best in specific time windows reveals its potential to have better predictive ability in specific situations. I will also further explore the potential of Random Forest under specific traffic flow conditions, comparing it to the CNN-LSTM model.

4.1.4 Errors in analysing results

The prediction errors and residuals of the four models (Random Forest, CNN-LSTM, SVM and LSTM) are analysed below, aiming to visualise the performance of the models through error distribution plots and residual plots, in order to better understand the strengths and weaknesses of each model:

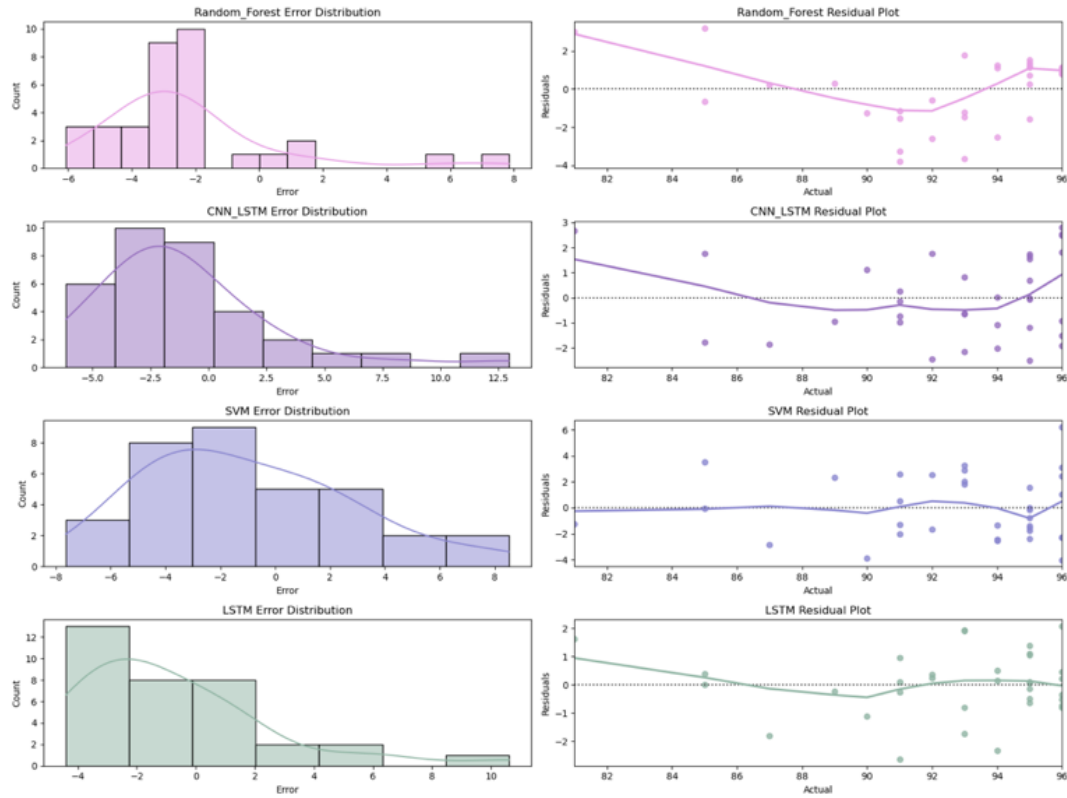


Figure 17: Error Distribution and Residual Analysis for four models.

For the error distribution diagram of the four models, CNN-LSTM model, and LSTM model both have a right-biased distribution trend, and the result error is divided into five right-biased distributions, indicating that the actual value may be overestimated in some cases. On the contrary, RF model shows a serious left-biased distribution trend,

indicating that most of the predicted value may underestimate the actual value and there are more negative errors. Although the SVM error is concentrated around zero and its error fluctuation is not large, the long tail on both sides indicates that there are serious prediction errors in some cases. At the same time, the error fluctuation range of CNN-LSTM model and LSTM model is small, and the error distribution of RF model is the most dispersed, which can also be seen that the two LSTM models have a certain stability for the results.

In the residual analysis, the residual of the RF model systematically decreases with the increase of the actual value, and the linear trend is obvious, which shows that the model may not capture some key information in the data well. For SVM, although its residuals show a tendency to be relatively flat and concentrated near the zero line, there are still some obvious outliers, which proves that although it is consistent in the overall prediction, the fitting degree for specific points is insufficient. The same problem also exists in CNN-LSTM model and LSTM model. There are large outliers at specific points, but both maintain low relative residual values, showing better overall prediction performance. CNN-LSTM model still shows better accuracy and stability in residual and error distribution graphs. The second is LSTM model, which has good stability. The Random Forest has certain adaptability in some cases, but it fluctuates greatly in general.

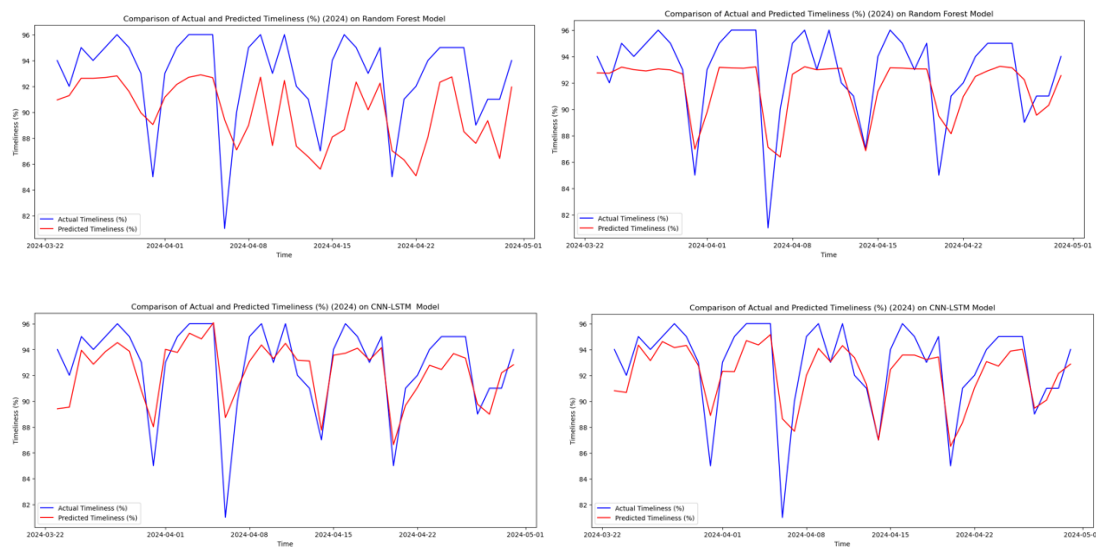
4.2 Modelling Optimisation Analysis

4.2.1 Impact of the lag algorithm on the model

In the analysis of the results of different time Windows (4.1.3), the model optimization is inspired, and it is found that the four models still have room for improvement in the time dependence and volatility of traffic flow data processing. Therefore, in order to solve this gap, we try to introduce the hysteresis feature function, which, as the name suggests, can provide the historical background of the data for the model and increase the medium - and long-term dependence of the data in the capture time series. The

following is the prediction result data of the four models after optimization.

	RF	CNN-LSTM	LSTM	SVM	P-value	T-statistic
RMSE	3.47	3.34	3.49	3.76	0.01499	0.0489
New RMSE	3.26	3.06	3.27	3.30		



The RMSE index clearly indicates that the performance of all four models significantly improved after incorporating the lag feature function. To further validate the significance of this optimization, a paired T-test was conducted, yielding a P-value of 0.01499 and a t statistic of 5.0489, confirming that this optimization method has a notable impact on model performance. Specifically, while the decline in RMSE values across the four models after optimization shows similar trends, the CNN-LSTM model continues to deliver the best predictive results. Although its decline is not the largest, it remains close to the second best.

To better illustrate the enhancement brought by the lag function, the CNN-LSTM model and the RF model were highlighted. The attached figure shows that the RF model's fit between predicted and actual values, as well as its ability to manage data fluctuations, improved significantly. Although the CNN-LSTM model did not show a substantial increase in accuracy, its capability to handle extreme values improved, offering

potential to address the model's previously noted shortcomings. The performance improvement observed with the lag feature function aligns with Kim's (2024) assertion that "Strategic selection of time windows and lag features is key to enhancing model performance." This study further substantiates this perspective.

4.2.2 Collaboration and repulsion of two hyperparameters

In this study, two main approaches were used to optimise the performance of the model: Lagged Features and Rolling Mean, where Lagged Features focus on short-term autocorrelation and predict the current data by taking into account the data from previous time points. Rolling Mean focuses on long term trends or fluctuations by smoothing the data to reduce the effect of noise. The combination of the two allows the model to capture both short-term fluctuations and long-term trends. However, in that study, it was found that the two optimisations may act in a repugnant way, especially in machine learning models (SVM will be taken as an example for illustration below), as the model may be able to capture detailed fluctuation information from lagged features, but the rolling average feature smooth out this fluctuation information, resulting in inconsistent signals received by the model. The model may struggle to learn the useful information contained in both features at the same time, thus affecting the prediction performance:

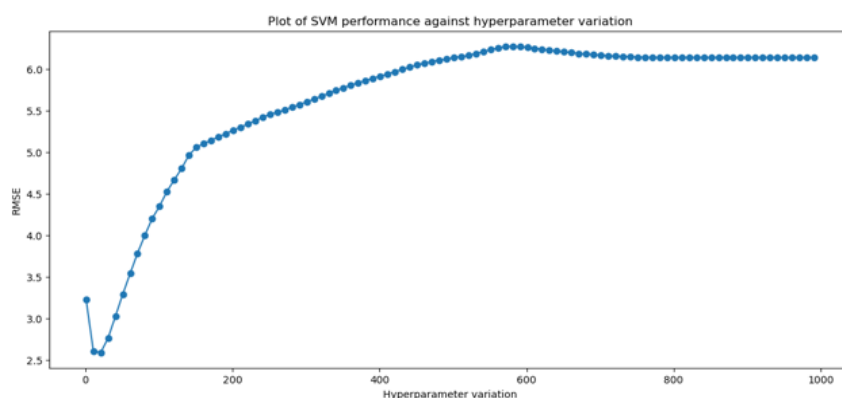
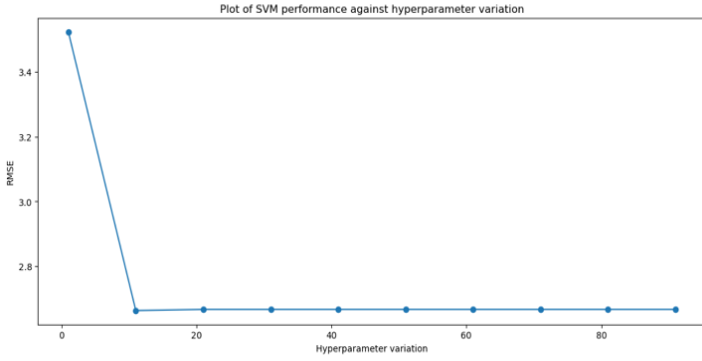
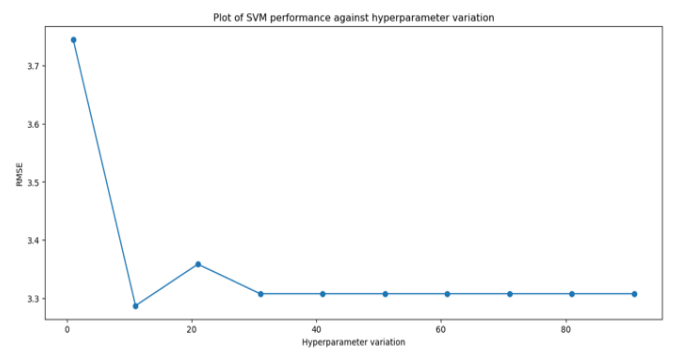


Figure 18: No hysteresis function was used. And the rolling average was 20.

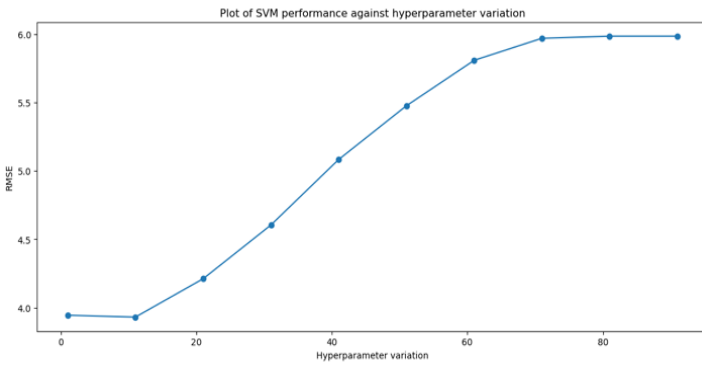
The following figure shows the fitness of the SVM for four different hyper-parameter cases in four:



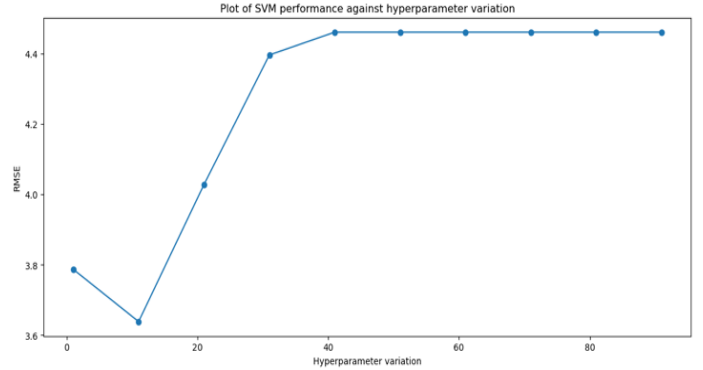
The lag function was 10 and the rolling average was 7.



The lag function was 10 and the rolling average was 10.



The lag function was 10 and the rolling average was 20.



The lag function was 20 and the rolling average was 20.

These plots show that the SVM exhibits better performance when the number of lagged features is 10 and the rolling average window is 7, indicating that the lagged features are effective in capturing short-term fluctuation information in a shorter time window, while the rolling average appropriately smooth out the data noise. However, as the rolling average window increases (e.g., from 7 to 20), the performance of the SVM gradually decreases, probably because a larger rolling average window eliminates too much short-term fluctuation information, resulting in the model not being able to fully utilise the key information provided by the lagged features. When the number of lagged features is 20 and the rolling average window is also 20, the performance of the model decreases significantly, indicating that in this case, the model is difficult to process the complex information brought by these two features at the same time, which affects the prediction accuracy.

In summary, lagged features and rolling averages have their unique roles in time series modelling, but the combination of the two needs to be used with caution. We found that

when lagged features and rolling average features are introduced into the model at the same time, they may have a repulsive effect, especially if the hyperparameters are not set properly. Therefore, in practice, it is crucial to choose the right number of lag features and rolling average window size to avoid the problem of signal inconsistency and to optimise the predictive performance of the model.

4.3 Extreme condition analysis

Based on the insights from section 4.1.3, further exploration of the Random Forest and CNN-LSTM model. To verify the accuracy of the models, data from January 15, 2024, to April 30, 2024, for model training, and used data from May 1, 2024, to May 14, 2024, for prediction. Additionally, to better highlight the differences between the two models and considering the minimal fluctuation in 'timeliness' during this period, the testing target was changed to 'travel time.' The following section demonstrates that when dealing with a dataset involving fewer factors, such as "travel time," "vehicle speed," and "timeliness," as in this study, the performance of the Random Forest surpasses that of the CNN-LSTM model.

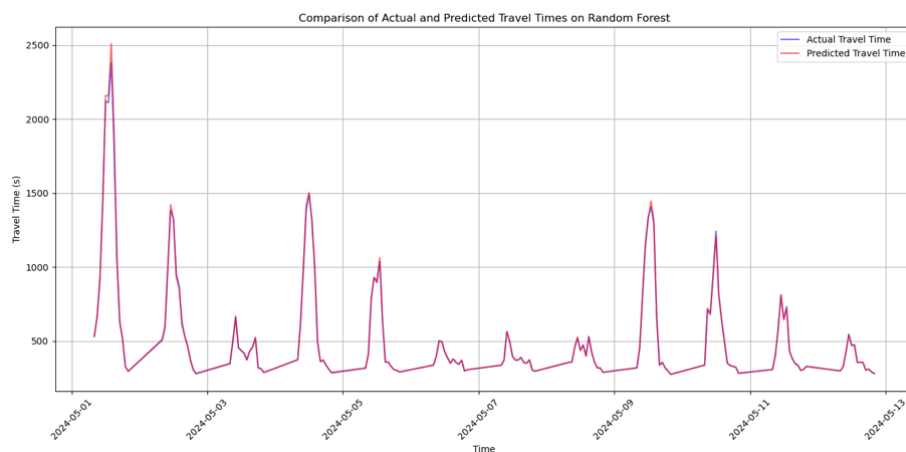


Figure 19: Random Forest predictions vs. actual travel time.

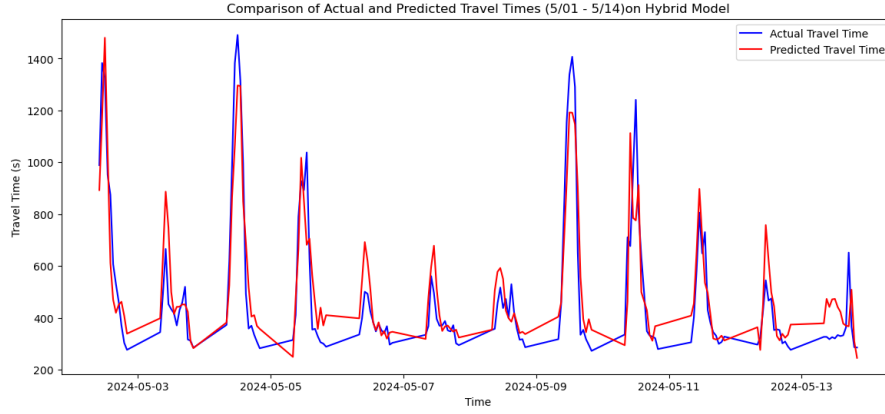


Figure 20: CNN-LSTM model predictions vs. actual Travel times.

Therefore, to evaluate the performance of the two models more specifically on different datasets and time series windows, the following section will employ external validation methods. External validation will be conducted by setting six different conditions, detailed as follows: in scenarios 1 to 3, the dataset includes travel time, vehicle speed, and timeliness, while in scenarios 4 to 6, weather conditions, temperature, and event occurrences are further added to the aforementioned dataset:

Scenario 1: Using the period from January 2024 15 to 30 April 2024 for model training, and data from 1 May 2024 to 14 May 2024 for testing.

Scenario 2: the training data was selected from 23 March 2024 to 30 April 2024, and the testing data from 1 May 2024 to 14 May 2024.

Scenario 3: the data from 23 March 2022 to 30 April 2024 for the longer time data as the training set, while the data from 1 May 2024 to 14 May 2024 was used for testing.

	RMSE		R2	
	CNN-LSTM	Random Forest	CNN-LSTM	Random Forest
Scenario 1	4.51	3.18	0.39	0.76
Scenario 2	5.2	3.6	0.65	0.69
Scenario 3	3.90	3.78	0.60	0.68
Scenario 4	2.98	4.32	0.66	0.32
Scenario 5	3.34	3.47	0.3	-0.02
Scenario 6	3.01	4.8	0.77	0.36

The RMSE of CNN-LSTM is generally higher than that of Random Forest in Scenario s 1-3, which may be since Random Forest is better able to capture key information and make effective predictions with fewer features. Random forest has higher R_2 values in most Scenario s, especially in Scenario 1, where the R_2 of Random Forest is 0.76, which is significantly higher than that of CNN-LSTM's 0.39, suggesting that Random Forest is more capable of explaining data variability with this simple combination of features; however, in Scenario s 4-6, weather and temperature information is added. Compared to the first three Scenario s, the CNN-LSTM showed a more significant improvement in performance, and the R_2 values of the Random Forest generally decreased with the addition of weather and temperature information, with a negative value (-0.02) especially in Scenario 5, which may indicate that the model overfitted or was not able to effectively account for the additional data variability when dealing with the increased complexity.

This finding highlights the need to further consider the endogeneity of the data when selecting a model. For Scenario 1-3, the selected data set only contains the three factors of "travel time", "vehicle speed" and "timeliness", but there are still some endogenous problems among the three factors. In other words, there may be a relationship between data with linear characteristics. In Scenario 4-6, when complete non-linear feature data is added, the performance of RF model will immediately decline dramatically, and even a situation of model underfitting may occur, while the performance of CNN-LSTM model will not only be improved, but also be improved. At the same time, it also maintains a certain stability. Although the performance of CNN-LSTM model in Scenario 1-3 is much lower than that of RF model, it also shows its potential in processing data with highly non-linear features in a more comprehensive way.

5. Discussion and conclusion

5.1 Discussion

In this study, the CNN-LSTM hybrid model was utilized to predict traffic flow, yielding effective results in processing the complex spatio-temporal patterns observed during the open season at Keukenhof Gardens. Various testing methods, data error analysis, and model optimization efforts indicate that this model outperforms traditional machine learning methods such as SVM and Random Forest. Compared to the LSTM model alone, the hybrid approach significantly enhances the ability to capture time series and nonlinear features within the dataset. This finding aligns with the results of Chen et al. (2016) and Lee & Min (2018), who emphasize the advantages of deep learning models in handling complex spatio-temporal data. The decision to combine CNN and LSTM was driven by the current trend towards hybrid models, addressing the limitations of a single LSTM in managing both spatial and temporal dependencies. A key advantage of CNN is its effectiveness in extracting periodic data features related to traffic flow, while LSTM is better suited to adapting to changes in traffic patterns during subsequent processing. The integration of CNN and LSTM into a hybrid model was aimed at capturing both spatial and temporal dependencies, which individual models often fail to achieve. This approach is similar to the strategy used by Cui et al. (2020), who successfully applied a comparable traffic prediction architecture in scenarios with incomplete data. In this study, the CNN layer effectively extracted features related to the daily and weekly cycles of traffic speed, while the LSTM layer processed these features over time, enabling the model to dynamically adapt to changing traffic patterns. Although the CNN-LSTM hybrid model demonstrates great potential in predicting traffic flow at Keukenhof Gardens during peak hours, some limitations remain:

1. In 4.1.3, the test of the mixed model in different time Windows shows that the model still has a high sensitivity to data learning in a short period of time. Although it can learn more historical data in a long window, it also means that the problem of gradient explosion is easy to occur in the learning process.

2. In addition, the hybrid model involves two neural network models, which also means that the code needs to be constantly debugging to find the current optimal parameter, but the parameter may change due to the difference of data sets each time.
3. Finally, the coordination relationship between the convolution layer of CNN and the pooling layer. The convolution layer extracts the local features of the time series through the filter, while the pooling layer will downsample these features to reduce the data dimension and prevent overfitting. This also means that if the collaboration between the two is not good, a lot of useful information can be lost.

So, there are still many ways to improve the potential of this model in future learning:

1. Further improve its sensitivity to traffic flow data by introducing further other mechanisms (such as attention mechanisms).
2. It is also possible to explore the addition of other advanced neural network structures (such as deformers or gated recursive unit GRUs) to the existing CNN-LSTM framework.
3. Try more optimization methods to adjust the hyperparameters of the model, not only limited to the LSTM model, but also optimize the CNN module, such as ResNet, VGG, etc.

5.2 Conclusion

The purpose of this study is to use the hybrid CNN-LSTM model to predict the traffic flow of Keukenhof Garden during the peak period by analogy with other traditional forecasting models. The results show that the CNN-LSTM model can learn complex hidden spatiotemporal patterns in traffic data, and in most cases, the performance of the hybrid model is better than other models. Based on its ability to learn spatial and temporal dependence, its prediction results are more accurate and robust, especially in dynamic environments such as tourist seasons. The results also show the sensitivity of CNN-LSTM model to time window selection. In other words, the longer the time window, the longer the sliding sequence, and by providing more historical context to improve the prediction accuracy, the performance and stability of the hybrid model will

become better and better. Unfortunately, on the other hand, the model still guarantees excellent stability and model fit when connected for a shorter time window. In addition, while the CNN-LSTM hybrid model is very effective when dealing with complex, high-dimensional data, it may not always be applicable when there are fewer data influences or more linear data. Finally, it is proved by experiments that adding hysteresis features to all nonlinear models is beneficial because it enhances the ability of these models to capture time dependencies, thus improving the overall prediction accuracy. All in all, this study proves that the CNN-LSTM hybrid model has great potential in traffic flow prediction in complex environment and has better performance than traditional nonlinear prediction model and single LSTM model.

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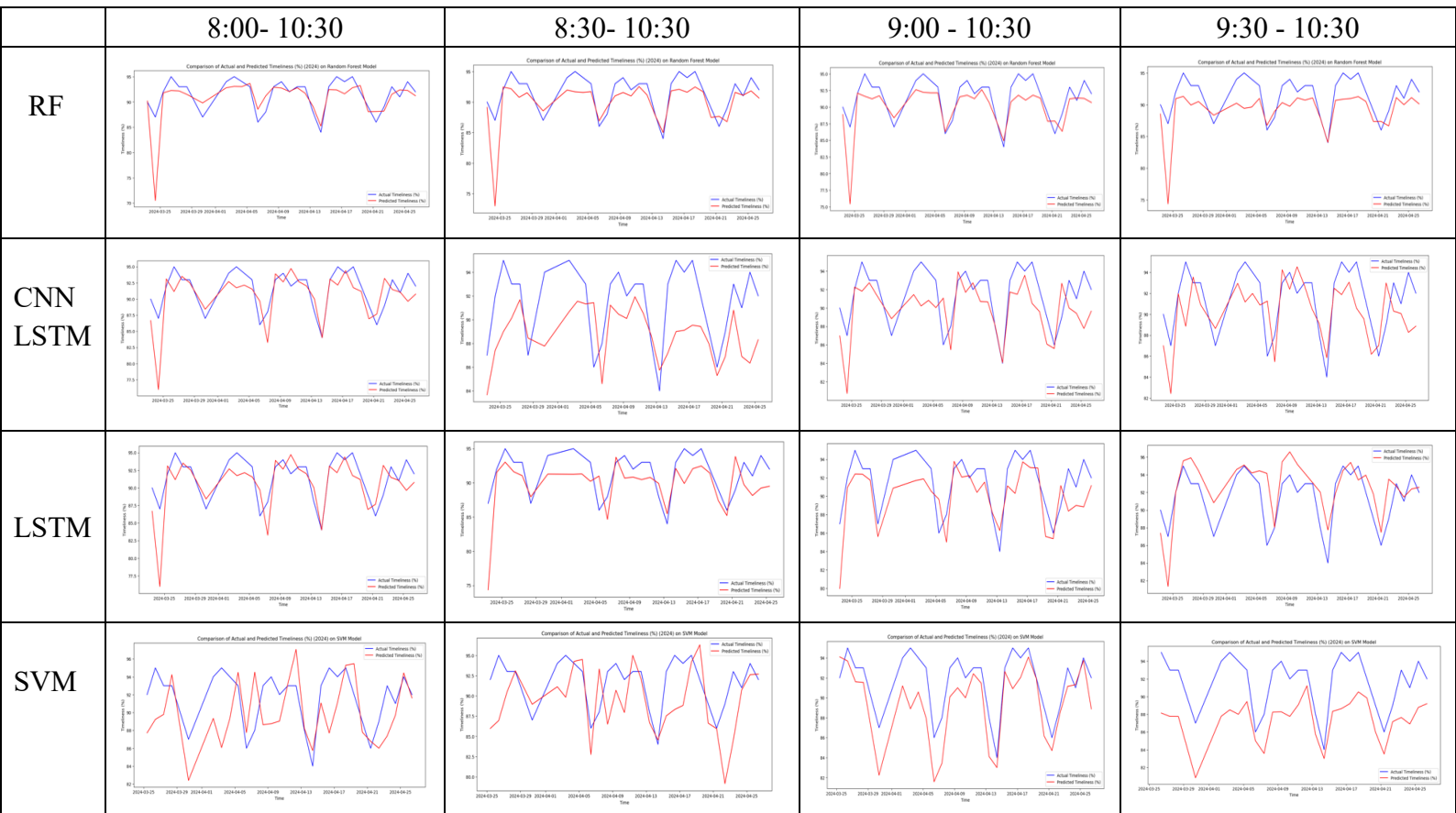
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7. Appendix

7.1 Here's a plot of the predicted values versus the true values for these four models over four-time windows:



7.2 Modeling code and data used in the study:

<https://github.com/ALLEN-2001/UCL-Dissertation>

Statement of Risk Assessment & Ethics Approval Requirements
Student Candidate Number: HFFF7 Student Name: Dayang Chen Student UCL Email Address: ZCZQDC1@ucl.ac.uk Supervisor Name: Manos Chaniotakis Supervisor UCL Email Address: m.chaniotakis@ucl.ac.uk Dissertation Research Proposal ☐ Title / Topic: Machine Learning-driven Transport network modelling ☐ Research Question(s) / Aims & Objectives: This study aims at exploring the representation of traffic dynamics using machine learning. The primary idea is that using the inputs and outputs (flows and ODs) we will be able to define a regression model that represents the dynamics of the simulator. Sensitivity analysis would have to be performed to understand the impact that network complexity has to the model defined. ☐ Data & source (specify all data to be used; if none, explain why): Not, use simulate data during processing ☐ Method(s): Machine Learning
I have read and understood Step A1 'Does the research require a Risk Assessment?' and: ☐ This planned research does NOT require a risk assessment.
I have read and understood Step A2 'Does the research require External research ethics approval?' and: ☐ This planned research does NOT require external ethics review.
External ethics approval is <i>not required</i> and I have read and understood Step A3 'Is the research Exempt from the need for ethics approval?' and: ☐ This planned research is NOT EXEMPT from the need for research ethics approval.
The research is <i>not exempt</i> from the need for ethics approval and I have read and understood Step A4 'Does the research require High Risk ethics approval?' and: ☐ This planned research is NOT deemed high risk.
The research is <i>not exempt</i> from the need for ethics approval, does not require high risk ethics approval and: I have read and understood Step A5 'Does the research require ESDA low risk ethics review for questions-based methods OR BSEER low risk ethics review for other methods?' and: ☐ This planned research requires BSEER low risk ethics approval (for other methods), which will be secured before data collection starts.
I confirm that: ☐ the information I have provided is accurate to the best of my knowledge. ☐ if the answers to any of these questions changes, I will go through this protocol again.

NEXT STEPS:

- ☐ **STUDENT:** Copy the text of the *completed* statement above into an email and email it to your supervisor.
- ☐ **SUPERVISOR:** Reply to the email confirming your approval of the completed statement, copying the Dissertation PGTA (r.alasmar@ucl.ac.uk). It is the student's responsibility to ensure that happens.
- ☐ **STUDENT:**
 - ☐ Include this A6 Statement as a Dissertation Appendix after you have BLACKED OUT YOUR NAME & EMAIL ADDRESS so the second marker can mark anonymously.

The Dissertation mark sheet asks the second marker whether this form was filled out correctly and, if not, what % mark deduction they recommend.